

7. Integration Techniques

7.4. Integration by Partial Fractions

7.4a Learning Objectives

- I can understand how to rewrite rational functions as a sum of partial fractions.
- I can understand when and how to integrate by partial fractions.

7.4.1. Motivation

How can we rewrite the function $f(x) = \frac{1}{x-4} - \frac{1}{x-5}$ ($f(x)$ equals 1 over $x - 4$ minus 1 over $x - 5$)?

$$\begin{aligned} f(x) &= \frac{1}{x-4} - \frac{1}{x-5} \\ &= \frac{x-5}{(x-4)(x-5)} - \frac{x-4}{(x-4)(x-5)} \\ &= \frac{x-5-x+4}{(x-4)(x-5)} \\ &= \frac{-1}{(x-4)(x-5)} \end{aligned}$$

$f(x)$ equals 1 over $x - 4$ minus 1 over $x - 5$ equals $x - 5$ over $(x - 4)(x - 5)$ minus $x - 4$ over $(x - 4)(x - 5)$ equals $x - 5 - x + 4$ over $(x - 4)(x - 5)$ equals -1 over $(x - 4)(x - 5)$

Which is easier to take the integral of:

$$\frac{1}{x-4} - \frac{1}{x-5} \quad \text{or} \quad \frac{-1}{(x-4)(x-5)}?$$

1 over $x - 4$ minus 1 over $x - 5$ or -1 over $(x - 4)(x - 5)$?

The difference on the left is easier to take the integral of.

So, if we are asked to integrate the expression on the right, it would be nice to rewrite it as the expression on the left first. This process of rewriting a rational function as a sum of “simpler” rational functions is called partial fraction decomposition.

7.4.2. Partial Fraction Decomposition

Theorem 7.4.1 Fundamental Theorem of Algebra

If $P(x)$ (P of x) is a polynomial with real coefficients, then $P(x)$ can be factored into linear $(ax + b)$ (a x plus b) and irreducible quadratic $(ax^2 + bx + c)$ (a x squared plus b x plus c) factors.

$$P(x) = \underbrace{(a_1x + b_1) \dots (a_nx + b_n)}_{\text{linear}} \underbrace{(A_1x^2 + B_1x + C_1) \dots (A_mx^2 + B_mx + C_m)}_{\text{quadratic}}$$

P of x equals a one x plus b one times dot dot dot times a n x plus b n times a one x squared plus b one x plus c one times dot dot dot a m x squared plus b m x plus c m .

Idea:

Say $f(x) = \frac{P(x)}{Q(x)}$ ($f(x)$ equals $P(x)$ over $Q(x)$) where $P(x)$ and $Q(x)$ are polynomials. Then, we can factor the denominator, $Q(x)$ ($Q(x)$), into its linear and quadratic terms:

$$Q(x) = (a_1x + b_1) \dots (a_nx + b_n)(A_1x^2 + B_1x + C_1) \dots (A_mx^2 + B_mx + C_m)$$

$Q(x)$ equals $(a_1x + b_1) \dots (a_nx + b_n)(A_1x^2 + B_1x + C_1) \dots (A_mx^2 + B_mx + C_m)$

Then we can write $f(x)$ ($f(x)$) as:

$$f(x) = S(x) + \frac{J_1}{a_1x + b_1} + \dots + \frac{J_n}{a_nx + b_n} + \frac{K_1}{A_1x^2 + B_1x + C_1} + \dots + \frac{J_m}{A_mx^2 + B_mx + C_m}$$

$f(x)$ equals $S(x)$ plus J_1 over $a_1x + b_1$ plus $\dots$ plus J_n over $a_nx + b_n$ plus K_1 over $A_1x^2 + B_1x + C_1$ plus $\dots$ plus J_m over $A_mx^2 + B_mx + C_m$

where $S(x)$ ($S(x)$) is a polynomial.

More precisely, the steps you can use are as follows.

Tip

General process for decomposing a rational function into its partial fractions.

1. If $f(x) = \frac{P(x)}{Q(x)}$ ($f(x)$ equals $P(x)$ over $Q(x)$) is not *proper* ($\deg P(x) \geq \deg Q(x)$ ($\deg$ of $P(x)$ is greater than or equal to $\deg$ of $Q(x)$)) then use polynomial long division to find

$$f(x) = S(x) + \frac{R(x)}{Q(x)}, \text{ with } \deg R(x) < \deg Q(x)$$

$f(x)$ equals $S(x)$ plus $R(x)$ over $Q(x)$, with $\deg$ of $R(x)$ is less than $\deg$ of $Q(x)$

2. Then, if $\frac{R(x)}{Q(x)}$ ($R(x)$ over $Q(x)$) cannot be integrated as is, factor $Q(x)$ ($Q(x)$) into linear and quadratic irreducible terms.
3. Split $\frac{R(x)}{Q(x)}$ ($R(x)$ over $Q(x)$) into its sum of partial fractions with unknown numerators.

Factor in $Q(x)$ ($Q(x)$)	Sum of Partial Fractions
$(ax + b)^m$ $(ax + b)^m$	$\frac{A_1}{ax + b} + \frac{A_2}{(ax + b)^2} + \dots + \frac{A_m}{(ax + b)^m}$ A sub 1 over a x plus b plus A sub 2 over (a x plus b) squared plus ... plus A sub m over (a x plus b) m
$(ax^2 + bx + c)^n$ $(ax^2 + bx + c)^n$	$\frac{B_1x + C_1}{ax^2 + bx + c} + \frac{B_2x + C_2}{(ax^2 + bx + c)^2} + \dots$ $+ \frac{B_mx + C_m}{(ax^2 + bx + c)^m}$ B sub 1 x plus C sub 1 over a x squared plus b x plus c plus B sub 2 x plus C sub 2 over (a x squared plus b x plus c) squared plus ... plus B sub m x plus C sub m over (a x squared plus b x plus c) m

4. Solve for the numerators.

1. Set original fraction equal to the sum of partial fractions.
2. Multiply both sides by $Q(x)$ ($Q(x)$).
3. Solve for unknown constants using a system of equations and/or plugging in some x (x) values.

Example 7.4.1

Use polynomial long division to rewrite

$$f(x) = \frac{x^4 + 3x^3 + 5x + 6}{x + 2}$$

$f(x)$ equals x to the power of 4 plus 3 x cubed plus 5 x plus 6 over x plus 2

then evaluate

$$\int f(x) dx$$

integral of $f(x)$ dx

Solution

So,

$$\frac{x^4 + 3x^3 + 5x + 6}{x + 2} = (x^3 + x^2 - 2x + 9) + \frac{-12}{x + 2}$$

x to the power of 4 plus 3 x cubed plus 5 x plus 6 over x plus 2 equals (x cubed plus x squared $- 2x$ plus 9) plus $- 12$ over x plus 2

Then, with a quick and small u (u)-sub,

$$\begin{aligned}\int \frac{x^4 + 3x^3 + 5x + 6}{x + 2} dx &= \int (x^3 + x^2 - 2x + 9) dx + \int \frac{-12}{x + 2} dx \\ &= \frac{x^4}{4} + \frac{x^3}{3} - x^2 + 9x - 12 \ln|x + 2| + C\end{aligned}$$

integral of x to the power of 4 plus 3 x cubed plus 5 x plus 6 over x plus 2 d x equals
integral of (x cubed plus x squared $- 2x$ plus 9) d x plus integral of $- 12$ over x plus 2 d
 x equals x to the power of 4 over 4 plus x cubed over 3 $- x$ squared plus 9 $x - 12 \ln$ of | x
plus 2 | plus C

Example 7.4.2

Set up the partial fraction decomposition for this rational function, but you do not need to find the coefficients of the terms in the numerators.

$$\frac{P(x)}{(x - 1)^3(2x - 3)(x^2 + 2x + 15)^2}$$

$P(x)$ over ($x - 1$) cubed ($2x - 3$) (x squared plus 2 x plus 15) squared

Solution

We first check the quadratic term is actually irreducible: since $b^2 - 4ac = 4 - 4(1)(15) < 0$ (b squared $- 4ac$ equals $4 - 4(1)(15)$ is less than 0), this is irreducible. Then, we have

$$\begin{aligned}&\frac{P(x)}{(x - 1)^3(2x - 3)(x^2 + 2x + 15)^2} = \\ &\frac{A}{x - 1} + \frac{B}{(x - 1)^2} + \frac{C}{(x - 1)^3} + \frac{D}{2x - 3} + \frac{Ex + F}{x^2 + 2x + 15} + \frac{Gx + H}{(x^2 + 2x + 15)^2}\end{aligned}$$

$P(x)$ over ($x - 1$) cubed ($2x - 3$) (x squared plus 2 x plus 15) squared equals A
over $x - 1$ plus B over ($x - 1$) squared plus C over ($x - 1$) cubed plus D over $2x - 3$
plus E x plus F over x squared plus 2 x plus 15 plus G x plus H over (x squared plus 2 x
plus 15) squared

7.4a Section Summary: